



Nursing Practice Guideline

Lumbar Puncture Management

Reference #: NPG17

Scope

Site	Service/Department/Unit	Disciplines
SCGOPHCG	Organisation-Wide	Nursing

Introduction

This guideline outlines the procedures and responsibilities of individuals and the health service organisation in relation to the lumbar puncture procedure. It ensures the delivery of a uniform, integrated approach with management that is consistent with best practice. This guideline does not cover the administration of intrathecal medication.

Risk Statement

Non-compliance with this guideline will:

Breach legislative requirements	☒	Impact on Patient Quality of Care	☒
Breach National/State/Hospital Policy	☒	Impact on Patient Safety	☒
Breach professional standards	☒	Misconduct	☐
Breach SCGOPHCG Mission & Values	☒	Staff Safety	☒
Other:	☐		

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1.1 Medical Requirements

Where clinically appropriate the Lumbar Puncture procedure requires written consent as per North Metropolitan Health Service Consent to Treatment Policy, and in compliance with the WA Health Consent to Treatment Policy (MP 0175/22).

Procedure is performed by a medical officer; who will also apply the dressing.

The medical officer should check:

- The need for and/or results of Computed Tomography (CT) before the procedure is performed.

- If the patient is on anticoagulation therapy or has abnormal coagulation studies.

Medical officer will advise regarding positioning, frequency of post procedural observations, and any post-procedural mobility restrictions.

Medical officer may prescribe medication to treat persistent post-procedural headache.

1.2 Safety Information

The Lumbar Puncture (LP) is a medical procedure which involves withdrawing cerebrospinal fluid by the insertion of a hollow spinal needle with a stylet in the lumbar subarachnoid space.^{1,2} The needle is placed in the interspace between L3/L4 or L4/L5 which is a safe distance from the end of the spinal cord which terminates at L1. Indications can be for diagnostic or therapeutic purposes.

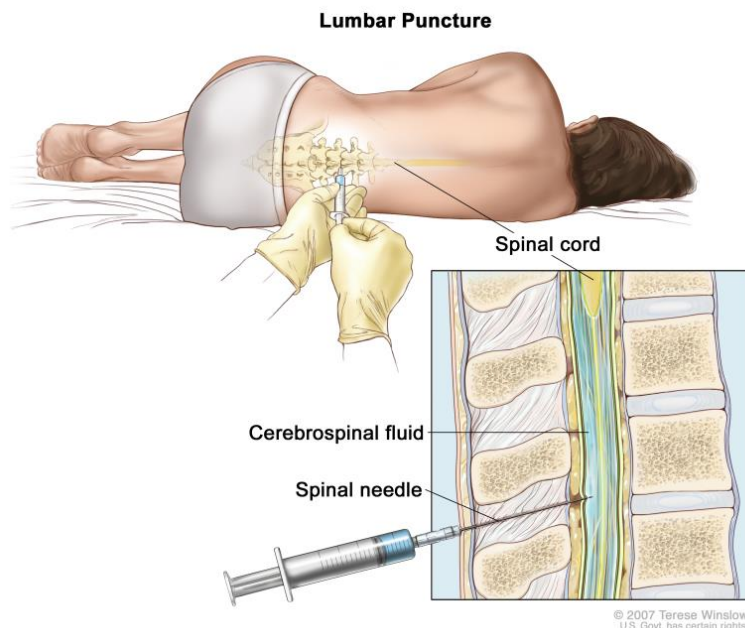


Figure 1 Illustration of Lumbar Spine³

Contraindications for performing a LP:

- Local skin infection over proposed lumbar puncture site or osseous infection at the site.¹
- Raised intracranial pressure (a head CT must be performed prior to the procedure)
- Abnormal blood coagulation¹.

Check if patient has received any anticoagulation therapy &/ or had coagulation profile within 24hrs.

Observe the five moments for hand hygiene, maintain surgical aseptic technique and maximal barrier precautions during the procedure. Follow the risk assessment outlined in SCGOPHCG Infection Prevention and Control Manual- Aseptic Technique Policy. This must include use of appropriate Personal Protective Equipment (PPE) to prevent contamination and mucosal or conjunctival splash injuries.

Skin antiseptic used must be allowed to dry to be completely effective.

Complications of a Lumbar Puncture:

- The most severe complication is trans tentorial herniation in the presence of raised intracerebral pressure (hence the requirement for pre-procedural head CT),

- Infection, haematoma and leakage from LP site,
- Bacterial meningitis has occurred following injection of material or insertion of a catheter into the epidural or subarachnoid spaces⁶
- Post LP headache is common (in about 1/3 of patients) and analgesia should be offered. This may be severe resulting in nausea and vomiting. Patients may also complain of back or leg ache.^{1,2.}

1.3 Nursing Management

Pre-procedure:

- Check and confirm consent for procedure has been completed.
- Check / document baseline neurological observations and core physiological observations.
- Explanation to the patient must include the need for aseptic procedure, positioning, the importance of remaining still during the procedure and the need to report tingling or other change of sensation in the lower limbs when the procedure commences.
- Review and address any need for analgesia and/ or antiemetic.
- Encourage the patient to void if required.
- Check proposed site, report signs of skin infection or other lesions to medical officer. Clip site, if needed, immediately before procedure.

Post-procedure:

- Ensure patient comprehends and follows directions provided by medical officer.
- Patient may mobilise unless contraindicated; lying down does not prevent headache^{2,3,5}
- Encourage oral fluid intake unless contraindicated³.
- Observe puncture site through transparent dressing, report ooze to medical officer.
- Assess and document neurological and core physiological observations at completion of procedure and 30 minutes post procedure; ongoing frequency and duration of observations is determined by medical officer.
- Report headache, neurological deterioration including deterioration of motor/sensory function to legs, or deterioration in core physiological observations to medical officer without delay.¹
- Lying down may help alleviate post-procedural (established) headache⁴.

1.4 Procedural Information

Requirements

Personal Protective Equipment (including surgical mask)

Drape pack

Dressing pack

Chlorhexidine 2% in Alcohol 70% solution (must be allowed to dry) Alternative if there is a known sensitivity to Chlorhexidine

1% Lignocaine 5 ml x 1 amps or alternative if there is a known sensitivity to Lignocaine

Adhesive, occlusive, transparent dressing

Needle 25g x 1½"

Needle 19g x 1½"

5ml syringe

Sterile gauze swabs

Spinal needles – 22g x 2

Spinal manometer

3-way tap
Sterile gloves
CSF collection tubes x 2
Absorbent sheet
Wet strength bag

Procedure

- Provide explanation and reassurance to the patient throughout.
- Perform hand hygiene.
- Position patient as requested by medical officer. Usually lying laterally with back flexed, pillow between legs and close to the edge of bed (see Figure 1).
- Remind the patient to remain still.
- Cover upper and lower body separately.
- Place an absorbent sheet under the lumbar area.
- Support aseptic technique for the procedure as needed.
- Assist with placement of transparent dressing over the site and monitor for any signs of leaking or infection.
- Assess and document neurological and core physiological observations as per post procedural Nursing Management.

Related Documents

Legislation:

- Health Practitioner Regulation National Law (WA) Act 2010
- Occupational Health, Safety and Welfare Act 1984.
- Nursing and Midwifery Board of Australia (NMBA) Code of Professional Conduct for Nurses March 2018
- Health Services Act 2016
- Work Health and Safety Act 2020

Policy:

- WA Health Consent to Treatment Policy MP 0175/22
- NMHS Consent to Treatment Policy
- NMHS Clinical Documentation Policy
- SCGOPHCG Infection Prevention and Control Manual, Aseptic Technique Policy
- SCGOPHCG Infection Prevention and Control Manual, Safe injection Practices Policy
- SCGOPHCG Clinical Handover - Communicating for Safety (HP#232)
- SCGH Patient Identification and Procedure Matching – Communicating for Safety

References

- (1) Hickey, J V. The Clinical Practice of Neurological and Neurosurgical Nursing (8th Ed). Lippincott, Williams & Wilkins, (2020)
- (2) Dougherty, L, Lister, S. The Royal Marsden Hospital Manual of Clinical Nursing Procedures (7th Ed). Wiley-Blackwell, (2008)
- (3) Adult Acute Lymphoblastic Leukemia Treatment (PDQ®) PDQ Adult Treatment Editorial Board. Published online: April 12, 2019. https://www.ncbi.nlm.nih.gov/books/NBK65844/figure/CDR0000257989_151/
- (4) Streak Gomersall J. Lumbar puncture: Clinician information. The Joanna Briggs Institute. 2013:1-4.
- (5) Arevalo-Rodriguez I et al. Posture and fluids for preventing post-dural puncture headache. Cochrane Database of Systematic Reviews 2016, Issue 3. Art. No.: CD009199. DOI: 10.1002/14651858.CD009199.pub3.
- (6) SCGOPHCG Infection Prevention and Control Manual, Safe injection Practices Policy. 2020

Authorisation & Version Control

Policy Sponsor	Nursing Practice Guideline Subcommittee				
Policy Contact	Nursing Practice Guideline Subcommittee Chair				
Date First Issued	Nov 1995	Last Reviewed:	March 2023	Review Date:	March 2026
Version No.	9.0				
Clinical Reviewer	P Ralphs CNS G52				
Research Reviewer	P Ralphs CNS G52				
Executive Sponsor	Executive Sponsor Nursing Practice Guideline Subcommittee				
Endorsed by	Nursing Practice Guideline Subcommittee			Date:	14/03/2023
Standards Applicable	NSQHS Standards (Version 2): 				

For full details regarding, development, review, consultation, approval, endorsement - contact the Policy Officer for the submission form

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